

Power III

MBI

Q

Which of the following schedules demonstrate a causal effect?

A

$Y_i(0)$	$Y_i(1)$
3	2
6	5
4	3
2	1
4	3
5	4

B

$Y_i(0)$	$Y_i(1)$
3	6
6	12
4	8
2	4
4	8
5	10

C

$Y_i(0)$	$Y_i(1)$
3	3
3	3
3	3
3	3
3	2
3	3

D

$Y_i(0)$	$Y_i(1)$
0	4
6	2
3	3
3	3
6	2
0	4

How to use your power wisely

1) Post-hoc power calculation

↳ generally unwise to plug in test stat as effect

2) Prospectively plan your Design + Analysis

a) Find n to get high power for a given effect size/ H_A

↳ estimate effect size using smallest meaningful (or previous literature).

b) At fixed n , know the range of effect sizes you'll be able to detect w/ high power.

c) At fixed n and effect size, which test statistic is highest power.

d) At fixed n , effect size, and statistic, which method of assignment is highest power.

Some Power Curves

[illegible]

Calculating $P_{H_A}(T \in R)$

What we need:

RBI

MBI

1. $R \hookrightarrow \hat{f}_{H_0}(T)$
 $\hookrightarrow \alpha = .05$

- a) Use data to make null schedule
- b) Randomly sample many partition and calculate t .

c) $R: [-\infty, q_{\alpha/2}] \cup [q_{1-\alpha/2}, \infty]$

2. $\hat{f}_{H_A}(T)$

- a) Use data to make alt sched.
- b) do 1.b)

3. $P_{H_A}(T \in R)$

- a) Calculate proportion of 2.b) that is in 1.c)

Power in Model Based Inference (MBI)

$$Y_i = \mu_j + \varepsilon_i \quad \varepsilon_i \sim N(0, \sigma^2)$$

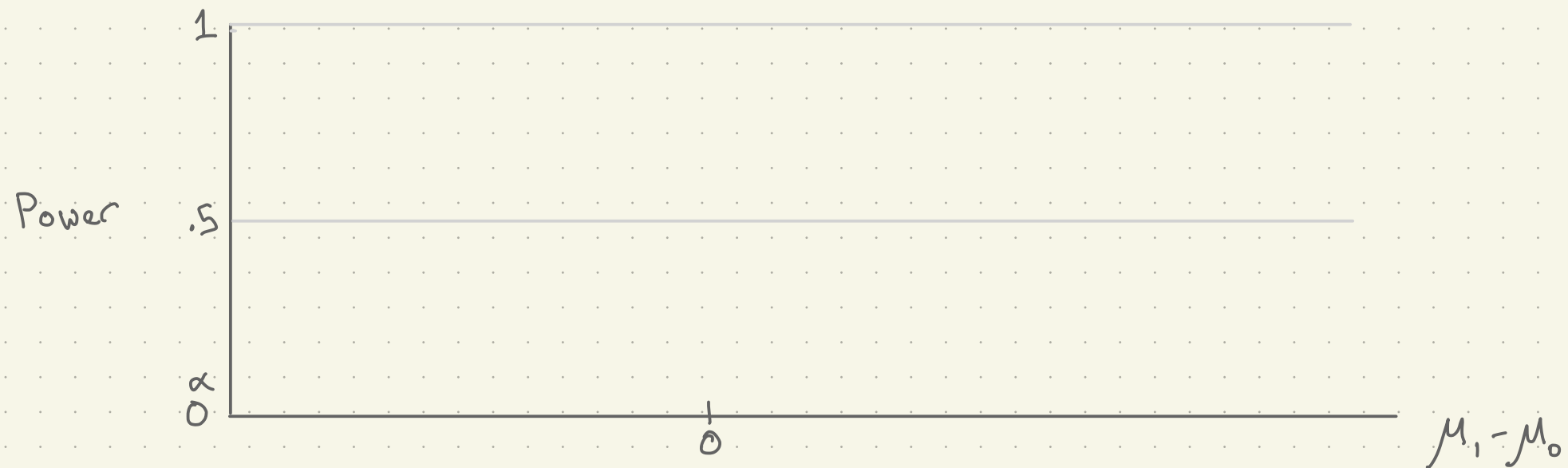
← analogs →

(RBI)

Ⓐ

$Y_i(0)$ $Y_i(1)$

3	2
6	5
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Calculating $P_{H_A}(T \in R)$

What we need:

RBI

$$1. R \rightarrow \hat{f}_{H_0}(T) \\ \rightarrow \alpha = .05$$

$$2. \hat{f}_{H_A}(T)$$

$$3. P_{H_A}(T \in R)$$

- a) Use data to make null schedule
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$$c) R: [-\infty, q_{\alpha/2}] \cup [q_{1-\alpha/2}, \infty]$$

- a) Use data to make alt sched.
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- a) Calculate proportion of 2.b) that is in 1.c)

MBI

$$a) \text{ If: } \cdot Y_i = \mu_j + \varepsilon_i, \varepsilon \sim N(0, \sigma^2) \\ \cdot T = \frac{\hat{Y}_1 - \hat{Y}_0}{SE}$$

$$\cdot H_0: \mu_1 - \mu_0 = 0$$

Then:

$$T_{H_0} \sim t(df = n_1 + n_0 - 2)$$

$$b) R: [-\infty, q_{\alpha/2}] \cup [q_{1-\alpha/2}, \infty]$$

↑ ↑
quantiles of T_{H_0}

$$a) \text{ If: } \cdot \text{same except} \\ \cdot H_A: \mu_1 - \mu_0 \neq 0$$

Then:

$$T_{H_A} \sim \text{noncentral } t(\delta = \mu_1 - \mu_0, df = n_1 + n_0 - 2)$$

$$a) \text{ Power} \\ F_{T_{H_A}}(q_{\alpha/2}) + [1 - F_{T_{H_A}}(q_{1-\alpha/2})]$$

↑ ↑
L side R side